



DURBAN UNIVERSITY OF TECHNOLOGY
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FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

2024

MID YEAR SUPPLEMENTARY EXAMINATION

INSTRUCTIONAL PROGRAMME : ADVANCE DIPLOMA ICT

INSTRUCTIONAL OFFERING : *Research Skills*

PAPER NUMBER : 1

SUBJECT CODE/S : RESK401

DATE : *2024/07/18*

DURATION : 3 HOURS

TIME : 09H00 - 12H00

TOTAL MARKS : 100

NUMBER OF PAGES : 4

EXAMINER/S : MISS LS MANQELE

MODERATOR/S : DR B MUTANGA

INSTRUCTIONS/REQUIREMENTS:-

1. This examination comprises of FOUR questions sections .
2. For QUESTION 1 AND QUESTION 2, use MCQ cards.
3. Use your pen or pencil. Make sure you write your Surname and initials with student number on the MCQ card
3. Use the mark allocations as a guide for the length of your answers.
4. On your MCQ cards: Question 2 asks if the statement is true or false. A is true and B is False.
Start question 2 from number 16 on your MCQ cards
5. Attempt ALL QUESTIONS.

Do not turn the page until permission is given

Question 1 [30]

1. This type of research is a research conducted for a problem that has not been clearly defined [2]
 - a. Explanatory research
 - b. Descriptive research
 - c. Exploratory research
 - d. Longitudinal research
2. One of the following statements is not the reason to review the literature [2]
 - a. To understand the state-of-the-art
 - b. To understand what work has already been done
 - c. To prevent setting up research for problems that have already been solved
 - d. To identify findings
3. The characteristics of popular sources of information include: [2]
 - a. Little to no citations to primary sources of information
 - b. Neutral and factual writing style
 - c. Use of field-specific vocabulary or nomenclature
 - d. Trustworthy publishers (IEEE, ACM, Springer)
4. This type of research focuses on exploring ideas and formulating a theory or hypothesis. [2]
 - a. Qualitative research
 - b. Mixed research
 - c. Quantitative research
 - d. None of the above
5. The type of survey that is used when the participants are selected for the sample using certain selection criteria. [2]
 - a. Cross-sectional survey
 - b. Panel survey
 - c. Cohort Survey
 - d. Trend survey
6. The research strategy that is used to determine the relationship between two aspects that are of interest in a research study. [2]
 - a. Survey
 - b. Design Research

- c. Experiment
- d. Case study

7. This type of sampling is used when the population is small or when there are no known characteristics of the population.? [2]

- a. Simple random sampling
- b. Quota sampling
- c. Snowball sampling
- d. Purposive sampling

8. This type of sampling Involves the selection of experts or those who demonstrate profound knowledge in the area of interest. [2]

- a. Expert sampling
- b. Quota sampling
- c. Stratified sampling
- d. Purposive sampling

9. Type of focus group where one or more client representatives participate in the discussion, either covertly or overtly [2]

- a. Client participant
- b. Correspondent moderator
- c. Mini focus group
- d. None of the above

10. The disadvantage of focus groups is: [2]

- a. A diverse set of responses
- b. A richer set of responses
- c. Strong personalities and outspoken individuals can dominate the group
- d. Likeminded group of participants

11. This type of data must belong to a category first before being assigned a numerical format. [2]

- a. Nominal
- b. Interval
- c. Categorical
- d. Ordinal

12. This example does not fall under the sensitive data category [2]

- a. Religion
- b. Race
- c. Political opinion
- d. Credit Card Pin

13. All data collection participants have a right to: [2]

- a) Not to participate
- b) Share the questionnaires with people they trust
- c) Expect payment for participation
- d) Not to sign the consent

14. One of the following options cannot be categorised as misconduct [2]

- a. Plagiarism
- b. Getting a consent
- c. Data fabrication
- d. Data manipulation

15. Data analysis is the process of:

- a. cleaning,
- b. Transforming
- c. Modelling data
- d. All of the above.

Question 2 [20]

1. Applied research contributes to an existing body of knowledge through original results. [2]
2. The objectives are the milestones that need to be achieved along the way. [2]
3. Predatory publishers are publications of very poor quality and have no reputation among academics and the scientific community. [2]
4. A: "Despite the boom, advances in ICT are seemingly almost always slightly behind developments in cybercrime" (Garvin, 2019: 21).

A is an example of an in-text referencing. [2]
5. Qualitative research is generally exploratory. [2]
6. True experiment involves the manipulation of the independent variables, and pre- and post-test of the dependent variable. [2]
7. Non-probability relies on the researcher's ability to select the sample. There is no element of random selection. [2]
8. Quantitative data can be generated from experiments, surveys and interviews, IoT sensor data, and drones. [2]
9. SPSS is one of the popular tools used for quantitative data analysis. [2]
10. Biometrics information is an example of personal data. [2]

Question 3: [20]

All questions are based on the following abstract:

Abstract

This paper reviews the literature on the impact of social media on the health of children and young people. Relevant papers were identified from Medline, Embase and PsycINFO databases. The studies

identified that the health impact of social media on children and young people was greatest on mental health and specifically in the areas of self-esteem and well-being, with related issues around cyberbullying and 'Facebook Depression', with an association between the use of social media and self-esteem and body image. However, it is difficult to determine the cause and effect, which is likely to be related to the nature of the young person. There is little work on the impact of social media on younger children. More research is needed to identify those most at risk of harm from social media and risk mitigation strategies to assist healthcare professionals in providing essential education for parents and young people.

1. What would be the most appropriate research method for this work [3]
2. Mention the portals used to identify the relevant articles [6]
3. What is the contribution of this study? [3]
4. What is the research problem for this study? [3]
5. Based on the abstract, what would be the potential topic for this study? [2]

Question 4 [30]

1. What is a direct quotation referencing method? [2]
2. What are the primary reasons for a researcher to use quantitative and qualitative research? [4]
3. What is the difference between pre-experiment and quasi-experiment? Give an example for each experiment type. [4]
4. Non-probability sampling is also known as random sampling, the most precise type of sampling. List and briefly describe any two types of probability sampling. [4]
5. One type of Inferential statistics is hypothesis testing. Write down any H_0 and H_1 hypothesis [4]
6. Give and explain three possible impacts of an unethical researcher in the research process [6]
7. Indicate whether the following data is personal, not personal or sensitive [6]
 - i. Sexuality
 - ii. Bank balance
 - iii. Timetable for RESK401

TOTAL: [100]